Bachelor of Science Honours in Computer Science
Second Year Group Project

DDROID

Smart Livestock Watering and Monitoring System

Software Requirements Specification

Group Project Requirements
Version 1.0 | 12 October 2026

Project team

Team member	Lead responsibility
Tapiwa Chigome	Project Manager
Fabilous Lashid	Software Engineer
Tanatswa Bernard Nyautore	Systems Engineer
Tinotenda Gozi	Network Engineer
Praise Muzvonda	Programmer 1
Nickson Gondwe	Programmer 2
Claris Chinanga	Test Programmer
Divine Pezari	Database Administrator
Lionel Tadiwanashe Tiugari	AI/ML Engineer
Keith Tawananyasha Mutabvuri	Chief Security Officer

Prepared by Tapiwa Chigome and Tanatswa Nyautore

Document control and contents

This Software Requirements Specification defines the DDROID livestock watering and monitoring system, its implementation scope, delivery plan and acceptance criteria. It provides the project team with a common baseline for developing and evaluating a supervised physical prototype.

Version	Date	Revision	Approval status
1.0	12 October 2026	Submission edition incorporating team review	Approved by project team

Contents

Section	Subject	Page
1	Introduction and project definition	3
2	Overall description	5
3	External interface requirements	7
4	System features	8
5	Other nonfunctional requirements	11
6	Other requirements and data design	13
7	Project delivery plan	14
7.2	Project timeline	15
7.3	Budget	16
7.4	Risk assessment	17
8	Throwaway prototype developed using vibe coding	18
9	Verification and requirements traceability	19
A and B	Glossary and analysis models	20
C	References	21

1 Introduction and project definition

1.1 Purpose

This document specifies DDROID version 1.0, a low-cost machine that automatically transfers water from an available supply container into a livestock drinking trough. A microcontroller reads water-level switches, controls a small pump and reports operating status. The first physical implementation is a supervised tabletop demonstrator with one supply container and one trough.

The project addresses routine water distribution at an existing water point. Its scope excludes water-source development, water-quality testing and animal-health diagnosis. The four-week implementation cycle will deliver a physical prototype demonstrating water-level sensing, controlled refill and local monitoring.

1.2 Document conventions

“Shall” identifies a mandatory and verifiable requirement. “Should” identifies a desirable enhancement. FR identifies a functional requirement; NFR identifies a nonfunctional requirement; OBJ identifies an objective; and AT identifies an acceptance test. All FR and NFR statements are mandatory unless explicitly labelled optional. Commissioning shall verify the specified thresholds against the actual hardware; any change shall follow section 6.4.

1.3 Intended audience

The intended readers are the lecturer or project supervisor, the DDROID project team, and farmers or livestock attendants involved in reviewing the design.

1.4 Product scope and project definition

1.4.1 Project title

DDROID Smart Livestock Watering and Monitoring System.

1.4.2 Problem definition

Reliable water access is a documented challenge in rural Zimbabwe. FAO describes difficulties affecting livestock in communities including Umzingwane [R3]. This project focuses on a narrower, testable problem: where a water supply already exists, an attendant may need to repeatedly check and refill a trough, while an empty supply tank or uncontrolled refill can go unnoticed.

DDROID addresses this problem through automatic level-based refill, a low-supply alarm and an event record. Week 1 user review will establish the frequency of current watering problems and the operational priorities of an intended user.

1.4.3 Aim

To design, build and evaluate an affordable computer-controlled livestock watering prototype that maintains a trough between defined low and high levels, alerts an attendant to supply or equipment faults, and continues local control without internet access.

1.4.4 Objectives

OBJ-01 By the end of Week 1, document the intended user's watering process, agree the project scope and review a throwaway interface and controller simulation with at least one intended user or approved representative.

OBJ-02 By the end of Week 2, connect the controller, level switches and low-voltage pump, and complete at least ten normal refill cycles without an observed overflow in the bench test.

OBJ-03 By the end of Week 3, implement supply protection, fault latching, a stop control, a maximum pump run timer, local status reporting and a timestamped event database.

OBJ-04 By the end of Week 4, execute the acceptance tests in section 9, resolve all failures affecting pump shutoff or access control, and deliver the agreed SRS, source files, wiring record, user guide and demonstration evidence.

OBJ-05 Compare ten supervised automatic refill cycles with ten manual refill cycles under the same bench conditions, recording refill time, observed overflow and attendant interventions. Report the measured results without assuming a percentage improvement.

1.4.5 Expected outcomes

- A documented throwaway software prototype showing the intended interface, control states and error responses.
- One working physical demonstrator that senses water levels, controls a pump and records alerts; a responsive local dashboard and a small SQLite database.
- A test report, bill of materials, wiring and calibration record, farmer or representative review notes, and a clear account of the system's limitations.

1.4.6 Justification

A watering machine is a practical fit for the team's livestock brief because it combines an observable physical task with software, communication, records and safety controls. A bench-scale design keeps procurement and testing manageable for second-year students while addressing a function that can later be investigated in a rural setting.

A mechanical float valve provides a useful comparison for cost and maintenance. DDROID adds fault alerts, operating records and control of a pumped supply. User review and bench testing will assess whether these benefits justify the additional equipment and maintenance requirements.

1.5 References

The structure of this SRS follows the coursework template [R1] and project brief [R2]. Full references are listed in Appendix C.

2 Overall description

2.1 Product perspective

DDROID is a new standalone water-point controller. It is a stationary computer-controlled machine, not a mobile robot. The microcontroller owns the pump-control loop; a local dashboard and database provide visibility and records. Dashboard, network or laptop failure must not remove the local pump interlocks.

Four float switches are proposed: trough low, trough high, supply low and a separate high-level cutoff above the normal high switch. A pump driver switches the low-voltage pump. A wired stop control and the independent high-level cutoff interrupt the pump-enable path separately from dashboard commands. Final component selection must confirm switch, driver and pump ratings.

2.2 Product functions

- Read the trough and supply states, refill between low and high marks, and stop on a fault.
- Show local pump state and alarms; allow a bounded manual fill and an intentional restart after a cleared fault.
- Send status and events to a local gateway, retain records, and export a report.
- Register a water point and its associated herd group without claiming to identify which individual animal drank.

2.3 User classes and characteristics

User class	Main actions	Access
Livestock attendant	Read status, acknowledge alarms, stop, reset a cleared fault, arm or request a manual fill	Operator
Farm owner or technician	Commission switches, set maximum run, manage accounts, view and export records	Administrator
Supervisor or observer	Inspect status and records during review	Viewer

Users may have limited technical experience. The interface shall use short labels, readable controls and explicit words such as LOW SUPPLY and PUMP OFF. Colour alone shall never communicate an alarm.

2.4 Operating environment

The bench machine uses an ESP32 development board, suitable 3.3 V digital inputs, a separately powered 5 V or 12 V DC pump and a rated driver. ESP32 supports local processing and wireless interfaces [R4]. A team laptop provides a local application service and SQLite storage over a short-range Wi-Fi network; the browser dashboard targets a phone and laptop.

The control loop shall work while the laptop or Wi-Fi is unavailable. A field installation would require a different enclosure, power assessment, water-compatible materials and supervised validation; these are outside the bench prototype's acceptance claim.

2.5 Design and implementation constraints

The baseline comprises one water point, one controller and a local service with no paid cloud dependency. Deterministic threshold rules shall govern actuation, and all pump commands shall remain subject to the safety interlocks. Machine-learning control is outside the baseline scope.

The implementation shall use ESP32 firmware in C or C++, a local Python service, SQLite, and an HTML/CSS/JavaScript dashboard. The implementation record shall identify framework and dependency versions. Browser access shall use HTTPS, and device telemetry and commands shall be authenticated. The wiring record shall identify the selected board and verified pin assignments.

The demonstration shall use low-voltage equipment with protected connections. The pump shall not be powered directly from a microcontroller pin. Selection and assembly must account for pump starting current, switching protection and separation of electronics from spills. Testing with live livestock is excluded from the four-week bench acceptance.

2.6 User documentation

Deliverable	Minimum contents
Quick start guide	Power on, verify water supply, arm, stop, read alarms and shut down.
Technician record	Actual components, circuit and pin mapping, switch positions, polarity, power ratings and calibration.
Fault guide	What each alarm means, checks before reset and when to leave the machine stopped.
Project evidence pack	Source, build instructions, prompt record, test report, user feedback and known limitations.

2.7 Assumptions and dependencies

- A safe existing water supply is available; neither borehole construction nor drought relief is part of the build.
- The team can borrow a laptop, basic tools and a phone, and obtain a suitable low-voltage power source. The budget identifies purchases separately.
- Components can be obtained during Week 1 and a supervisor can review the proposed design. Procurement delay triggers the fallback in the risk register.
- A farmer, livestock attendant or supervisor-approved representative will be available for prototype review. The review record will identify each participant's role.
- The implementation period is four working weeks from the agreed project start, as detailed in section 7.2.

3 External interface requirements

3.1 User interfaces

The dashboard shall show the water-point name, trough state LOW/ADEQUATE/FULL, supply state AVAILABLE/LOW, pump ON/OFF, operating mode, active alarm and last update time. It shall provide Stop, Arm automatic, Manual fill, Acknowledge, Reset fault and Export controls according to role. Confirmation shall be required before arming or starting a manual fill. Invalid actions shall return an explanation.

The physical panel shall provide a wired stop control, power indication, pump indication and an audible or visible alarm. The browser shall remain usable at 360 CSS pixels width without horizontal scrolling in its primary controls. History tables may scroll horizontally. No continuous percentage or litre value shall be shown unless a calibrated measurement device is added to a later scope.

3.2 Hardware interfaces

Interface	Information or action	Required behavior
Trough low and high switches	Digital wet or dry state	Sample every second; debounce for two consecutive samples. High wet while low dry is inconsistent.
Supply low switch	Available or insufficient supply	Low supply inhibits the pump and raises a latched alarm.
Independent high cutoff and stop	Separate pump-enable interruption	Remove pump drive independently of application and network code.
Driver and DC pump	ON or OFF output	OFF by default at startup, reset and loss of controller power.
Local indicators	State and alarm output	Readable or audible without a connected phone.

3.3 Software interfaces

The local service shall store accounts, water points, herd groups, readings, events and commands in SQLite. The browser shall read status and history from the service and send operator commands only through authenticated endpoints. All database writes shall use parameterised statements and transactions. The controller shall accept only recognised command types.

3.4 Communications interfaces

Use a private local Wi-Fi network. Telemetry shall include device_id, boot_id, sequence, uptime_ms, observed_at when available, low_wet, high_wet, source_wet, pump_state, mode and fault_code. A status message shall be sent at least every five seconds. If no message arrives for 15 seconds, the dashboard shall label the data STALE and disable remote starts.

A command shall include command_id, type, actor and an expiry no more than five seconds after issue. The controller shall reject expired, duplicate or unauthenticated commands; it shall return accepted/rejected with a reason. Start commands shall never be replayed after reconnection. Local safety logic always takes precedence.

4 System features

4.1 Water level monitoring

Priority: mandatory. When the system starts or a level switch changes, the controller validates inputs and updates the reported state. Input failure shall not be interpreted as a request to run the pump.

FR-01 The controller shall sample the trough low, trough high and supply switches at least once per second, and accept a changed normal level only after two consecutive consistent samples.

FR-02 The controller shall classify trough state as LOW when both normal trough switches are dry, ADEQUATE when only the low switch is wet, and FULL when both are wet. High wet with low dry shall raise SENSOR FAULT.

FR-03 The controller shall show its operating state and pump state locally and include the latest validated readings in each telemetry message. A raw independent cutoff shall act without waiting for software debounce.

4.2 Automatic refill control

Priority: mandatory. An operator checks the supply and arms automatic mode. A low trough initiates refill. The pump continues through the middle band and stops at the high switch, avoiding repeated switching around the low mark.

FR-04 The controller shall start in STOPPED with the pump OFF after power-up or restart. Automatic refill shall require an explicit authorised arm action and valid, safe inputs.

FR-05 While armed and with adequate supply, the controller shall start a refill when the validated trough state is LOW and continue until FULL, an interlock, an operator stop or the maximum run timer stops it.

FR-06 After a normal full stop, automatic mode shall remain armed in IDLE and shall not start another refill until LOW is detected. A drop from FULL to ADEQUATE alone shall not start the pump.

FR-07 Each refill shall have a continuous run timer. At the configured maximum, the controller shall turn the pump OFF, latch PUMP TIMEOUT and require fault reset before rearming. The initial bench setting shall be 60 seconds.

Acceptance focus

AT-01 verifies normal cycling and hysteresis. AT-02 verifies startup and stop behavior. AT-04 verifies timer shutoff. Float positions and the run limit must be set for the actual container geometry and pump flow before the first wet test.

A “maximum run” is a protection timer, not a water-use measurement. With no flow meter, the system cannot report litres dispensed, detect every leak or prove that the pump actually moved water. A blocked pipe is expected to cause the timer to expire.

4.3 Fault detection and alarms

Priority: mandatory. Low supply, inconsistent sensor states, independent high-level cutoff or timeout stops the pump and creates an alarm. Acknowledgement records that an operator has seen it; it does not release the interlock.

FR-08 The controller shall inhibit or stop the pump when supply is LOW, latch LOW SUPPLY and keep the pump OFF after supply returns until an operator resets the cleared fault and explicitly rearms.

FR-09 The controller shall inhibit or stop the pump on SENSOR FAULT or an independent high-level cutoff, latch the cause and record the event. The wired cutoff shall interrupt pump enable even if the software is unresponsive.

FR-10 The system shall allow an operator to acknowledge an alarm without clearing its cause, enabling the pump or removing the active alarm from the display.

FR-11 The controller shall reject fault reset while the triggering unsafe input remains active. A successful reset shall leave the machine STOPPED with the pump OFF.

4.4 Manual control and operator stop

Priority: mandatory. Manual operation assists a supervised demonstration or maintenance check. It remains subject to every supply, high-level and fault interlock.

FR-12 An authorised operator shall be able to request a manual fill only from STOPPED with safe inputs. It shall end after at most 15 seconds, or sooner on FULL, a fault or Stop, and shall leave automatic mode disarmed.

FR-13 A local wired stop shall remove pump drive irrespective of login, dashboard, network or application state. A valid remote Stop shall also disarm automatic mode. Neither stop shall cause an automatic restart when released.

4.5 Configuration

FR-14 Only an administrator shall change the automatic maximum run setting, and only while STOPPED. The accepted range shall be whole seconds from 10 to 120; invalid input shall leave the previous setting unchanged.

FR-15 The system shall log every configuration change with actor, time, previous value and new value. Approved settings shall survive a normal controller restart. Changing a setting shall never arm the pump.

Normal low and high limits are established by physical switch positions. Moving these switches is a technician commissioning action requiring a new calibration record, not a dashboard percentage setting.

4.6 Records and reports

FR-16 An administrator shall create and update a water-point record and its herd-group name and head count. The system shall reject a negative or noninteger count. No individual-animal drinking measurement shall be inferred.

FR-17 The local service shall store each received telemetry item and each state transition, alarm, acknowledgement and command outcome with its device identifier and sequence. Duplicate event deliveries shall not create duplicate records.

FR-18 An authorised user shall filter events by water point and date and export a CSV containing time, device, actor where applicable, event type and result. Empty results shall produce a valid header-only CSV.

4.7 Access control

FR-19 The service shall authenticate users and enforce Administrator, Operator and Viewer permissions on the server. Viewer requests to control the pump or change configuration shall be rejected even if a direct request bypasses the interface.

FR-20 An administrator shall create or disable user accounts. Disabling an account shall invalidate its active sessions. Failed login and denied control attempts shall be recorded without passwords or tokens.

4.8 Connectivity and recovery

FR-21 The controller shall continue local automatic control and enforce all interlocks when the gateway or Wi-Fi link fails. The dashboard shall label data STALE after 15 seconds without a status message and disable remote starts.

FR-22 The controller shall retain at least the latest 100 critical events during disconnection in a bounded local store and synchronise them after reconnection using boot_id and sequence for deduplication. It shall record any buffer overflow and telemetry gap.

FR-23 The controller shall reject expired, repeated or unauthenticated start commands. The service shall display a command as executed only after the controller acknowledges its outcome; on timeout it shall display UNKNOWN and require a fresh status check.

4.9 Optional future features

Solar or battery power, SMS alerts, additional water points and a Shona interface are optional future extensions. The AI/ML Engineer may investigate refill-pattern summaries using labelled synthetic data. Disease prediction, theft detection and predictive performance are outside the baseline scope. Optional work shall follow completion of the mandatory demonstrator.

5 Other nonfunctional requirements

5.1 Performance requirements

NFR-01 Under the normal one-second sampling cycle, a software-detected unsafe input shall result in pump OFF within three seconds, including input confirmation. Bench tests shall measure ten repetitions for each tested interlock.

NFR-02 On the local test network with one controller and three browser clients, the dashboard shall display a new received status within two seconds of service receipt. The controller shall send status at least every five seconds.

NFR-03 The system shall complete 20 normal bench refill cycles and a two-hour supervised mixed-scenario run without an uncommanded pump start or uncontrolled overflow. Any failure shall be corrected and the affected test repeated.

NFR-04 A history query covering 10,000 seeded records shall return its first result page within two seconds on the team’s recorded test laptop. Record the laptop specification and timing method in the test report.

5.2 Safety requirements

NFR-05 The pump output shall default to OFF at boot, watchdog reset and controller power loss. A hardware bias or equivalent verified circuit shall prevent a floating control pin from enabling the pump.

NFR-06 The wired stop and independent high-level cutoff shall remove pump drive within one second in ten bench tests, including a test with application processing deliberately halted. Reset or release shall not itself start the pump.

NFR-07 The assembled demonstrator shall use a separately fused low-voltage pump supply, a driver rated for the measured or specified starting current, suitable suppression and insulated connections separated from water. A supervisor shall review the wiring before wet testing.

NFR-08 Wet tests shall be supervised, use small containers within spill containment and retain a manual means to disconnect power. The team shall not connect the demonstrator to a live livestock water supply before a separate site and safety review.

Safety limits and residual risk

A float can stick, a wire can break and a switching device can fail closed. Inconsistent-state checks and a run timer do not detect every failure. The independent cutoff must therefore interrupt the pump power or enable path through appropriately rated hardware, rather than merely provide a second software input. A manual isolation point remains necessary.

Acceptance applies to a supervised bench demonstrator. Any deployment at a livestock water point requires a separate site assessment and field validation.

5.3 Security requirements

NFR-09 The service shall hash account passwords using a maintained Argon2id implementation with per-password salts and parameters benchmarked on the gateway. It shall not store or log plaintext passwords. Password handling shall follow OWASP guidance [R5].

NFR-10 Browser and device service traffic shall use authenticated encrypted connections, with device credentials separated from user accounts. No credential shall be committed in shared source. Invalid device identity shall be rejected and logged.

NFR-11 After five failed account logins within 15 minutes, the service shall block further attempts for 15 minutes and record the event. User sessions shall expire after 15 minutes of inactivity; pump operation remains subject to local controls.

NFR-12 All externally supplied fields shall be checked for type, permitted values and length before use. SQL queries shall be parameterised. Tests shall verify rejection of unauthorised commands, expired commands and malformed telemetry.

5.4 Software quality attributes

NFR-13 At least four of five approved representative users shall identify pump state, locate Stop and acknowledge an alarm within two minutes after a five-minute introduction, without assistance. The test record shall identify participant roles and task results.

NFR-14 The dashboard shall present text as well as colour for alarms and be operable by keyboard. Primary actions shall remain usable at 360 CSS pixels width and at a standard laptop width.

NFR-15 Source shall separate control logic, device drivers, service/database code and interface code. The repository shall include setup instructions, pinned dependencies, sample data without private identifiers, and a repeatable set of core control tests.

NFR-16 The local service shall retain telemetry for at least 30 days and critical events for at least 90 days while storage is available. A daily backup and a demonstrated restore shall recover all committed records in the last successful backup; storage failure shall be visible and shall not disable local pump interlocks.

5.5 Business rules

- Local safety interlocks take precedence over remote or manual commands. Acknowledgement is never permission to ignore a fault.
- Only an administrator may manage accounts and protected settings; an operator may operate an assigned water point; a viewer may only inspect and export permitted records.
- Only one assigned herd group is required per water point in this release. Counts describe the herd register and do not measure water intake.
- Field pilots and price offers require a separate team decision after the bench prototype and user review.

6 Other requirements and data design

6.1 Data requirements

The database shall use stable internal identifiers, foreign keys and transactions. Store times in UTC and display them in Africa/Harare for the coursework deployment. When a controller lacks a trusted wall clock, retain boot_id and uptime_ms, mark its time as unsynchronised and record gateway receipt time separately.

Entity	Core fields	Rules and relationship
User	user_id, username, password_hash, role, enabled	Unique username; no plaintext password.
WaterPoint	water_point_id, name, device_id, herd_id	Unique device ID; one active herd assignment.
HerdGroup	herd_id, name, head_count	Count is an integer of zero or more; no health inference.
Reading	device_id, boot_id, seq, observed_at, received_at, switch states, pump, mode	Unique device + boot + sequence; index receipt time.
Event	event_id, device_id, boot_id, seq, type, severity, actor, acknowledgement	Preserve original cause and later acknowledgement separately.
Command	command_id, actor, device_id, type, issued_at, expires_at, outcome	Unique command ID; executed, rejected or unknown outcome.
SettingChange	change_id, actor, device_id, old_value, new_value, time	Append audit entry with each accepted setting update.

6.2 Data integrity and retention

The service shall reject missing required fields and invalid switch values without crashing or updating the last valid state. An interrupted transaction shall not leave a partial event. Readings may be rotated after the retention period; critical events and audit data require an export or backup before deletion. A full event buffer must report the loss of older entries.

Daily backups belong to the local gateway. Offline control-device buffering protects only a limited event history; it does not promise a complete record of every telemetry sample during long disconnections or power loss. Recovery tests shall explicitly report gaps.

6.3 Privacy and research ethics

The prototype shall use synthetic herd names and records. Collect only necessary contact information for voluntary interviews, obtain consent and keep interview notes separate from operational data. Avoid publishing precise farm locations or identifying details in screenshots. A field pilot shall require permission from the owner and the supervisor.

6.4 Baseline and change control

Tapiwa Chigome maintains the change log. A proposed change records the affected requirement, reason, owner, time and cost impact, and test impact. Safety-related changes require the Systems Engineer and Test Programmer to review the design before team approval. Update this SRS version and the traceability matrix together.

7 Project delivery plan

7.1 Business model

The target customers are smallholder livestock farmers with an existing stored-water source, farmer groups managing a shared water point, and demonstration or training farms. The value proposition is convenient refill with local fault visibility and records, without a compulsory internet subscription. User review will assess willingness to pay and maintenance requirements.

Element	Proposed approach
Customer problem	Repeated manual checking, unnoticed supply depletion and uncontrolled refill at a managed water point.
Product and value	A locally serviceable controller kit, installation, commissioning and a simple local dashboard.
Revenue	One-time hardware and installation fee; optional maintenance visits and replacement parts. No mandatory cloud fee for baseline operation.
Channels	Direct demonstrations, farmer groups and agricultural support contacts, subject to their willingness to participate.
Main costs	Components, assembly, enclosure, installation travel, training, warranty returns and maintenance time.
Prospective partners	Livestock farmers, the project supervisor, component suppliers and a technician able to support future field trials.

Commercial feasibility test

Week 1 user interviews will examine the current watering process, inspection frequency, recurring failures and available repair support. The review will compare DDROID with a mechanical float valve and existing manual practice. Week 4 feedback will assess the value of the demonstrated functions against the anticipated equipment and maintenance costs.

For business planning, an assumed production and installation cost of USD 150 and a selling price of USD 200 would yield a contribution of USD 50 per unit, or 25% of sales, before overheads, warranty provision and taxes. These assumptions require supplier and customer validation. The coursework budget in section 7.3 covers one bench prototype.

Team contribution and accountability

The role on the cover identifies the lead, not the only contributor. All ten members participate in scoping, building and demonstrating. Each lead keeps a short task record and reviews another member’s work. A shared Kanban board uses To Do, In Progress, Testing and Done; a task is Done only when its evidence is linked.

7.2 Project timeline

The project will run for four working weeks, comprising 20 working days from the agreed start date. The physical demonstrator and acceptance evidence are due in Week 4. Subsequent semester activities may cover presentation and approved improvements.

Period	Work and lead owners	Milestone and evidence
Week 1 Days 1 to 5	Tapiwa: scope and task board. Tanatswa: hardware concept. Fabilous and Nickson: throwaway simulation review. Lionel: data and AI feasibility. All members: user review.	Agreed challenge; reviewed TP 1.0; updated SRS; decision log; components and supplier quotations. Archive throwaway code after learning is recorded.
Week 2 Days 6 to 10	Praise: device firmware. Tanatswa: sensors, driver and cutoff integration. Tinotenda: local network and message contract. Keith: access and safety review.	Wiring record and calibration; ten normal refill cycles; startup OFF, stop, low-supply and timer checks.
Week 3 Days 11 to 15	Nickson: dashboard. Divine: schema, records and backup. Fabilous: service integration. Tinotenda and Keith: authenticated communication and access tests.	Dashboard and machine integrated; event export; stale-data response; offline control; account and command checks.
Week 4 Days 16 to 20	Claris: acceptance and regression tests. Lionel: results analysis. Tapiwa: evidence and presentation. All members: defect resolution and demonstration.	20-cycle and two-hour tests; manual comparison; usability review; final SRS, source, report, user guide and physical demonstration.

Dependencies and review gates

Scope confirmation and component availability form the Week 1 review gate. Wiring approval is required before wet testing, and stable controller operation is required before dashboard control is enabled. The Week 4 demonstration requires all mandatory pump-stop tests to pass.

Component delays shall trigger a revised procurement and demonstration schedule agreed with the supervisor. Requirements review and simulation may continue while hardware is unavailable. Optional analytics, SMS and solar work shall be deferred before changes to the mandatory scope are considered.

Progress reporting

The Project Manager checks the task board twice per week. Every lead reports completed evidence, current blocker and next task. Claris maintains the defect list with severity, reproduction steps, owner, fix and retest result. The team keeps one agreed document version for all member submissions.

7.3 Budget

The estimated cash budget for one bench demonstrator is USD 169.20, including a 20% contingency. The estimates below are in United States dollars, dated 12 October 2026. Procurement shall confirm component specifications, availability, delivery charges and supplier prices before purchase.

Item	Qty	Unit USD	Total USD
ESP32 development board	1	15.00	15.00
Float switches including independent high cutoff	4	4.00	16.00
Small low-voltage DC water pump	1	12.00	12.00
Rated pump driver and switching protection	1	8.00	8.00
Wired stop and isolation components	1	7.00	7.00
Tubing and fittings	1	6.00	6.00
Supply and trough containers	2	6.00	12.00
DC pump power adapter	1	8.00	8.00
Voltage regulator or buck module	1	5.00	5.00
Fuse, holder and protected wiring	1	6.00	6.00
Electronics enclosure	1	10.00	10.00
Alarm indicators or buzzer	1	3.00	3.00
Prototype board and connectors	1	7.00	7.00
Local transport and sourcing allowance	1	15.00	15.00
Development data allowance	1	6.00	6.00
Printing and demonstration materials	1	5.00	5.00
Subtotal			141.00
Contingency at 20 percent			28.20
Total estimated cash budget			169.20

An equal allocation across the ten team members would be USD 16.92 per member.

Borrowed or already available: laptop, phone, basic hand tools and an internet-capable connection. Student labour is an in-kind contribution. Solar panels, batteries, field-rated vessels, GSM subscriptions and commercial manufacture are excluded. The contingency covers small substitutions and procurement variation; approve any larger overrun before purchase.

7.4 Risk assessment

Likelihood and impact are scored from 1 to 3: 1 low, 2 medium and 3 high. The risk score is likelihood × impact: 1–2 low, 3–4 medium and 6–9 high. The team shall reassess residual risk after implementing and verifying each mitigation.

ID	Risk	L × I	Mitigation and response	Owner
R-01	Switch sticks or reads incorrectly	2 × 3 = 6	Debounce, consistency check, independent high cutoff and timeout; test disconnected and stuck conditions.	Tanatswa / Claris
R-02	Pump runs dry or overfills	2 × 3 = 6	Supply interlock, rated independent cutoff, wired stop, spill tray and supervised wet tests.	Praise / Tanatswa
R-03	Power interruption or damaged switching hardware	2 × 3 = 6	Default OFF circuit, fuse and isolation; restart disarmed; test power removal and halted software.	Tanatswa
R-04	Wi-Fi or gateway fails	3 × 2 = 6	Keep local control independent; stale flag, bounded event buffer and no command replay.	Tinotenda / Divine
R-05	Unauthorised or malformed command	2 × 3 = 6	Server permissions, device authentication, encrypted transport, expiry and input validation.	Keith / Fabilous
R-06	Parts delayed or costs rise	2 × 2 = 4	Obtain quotes in Week 1; agree equivalent parts; use contingency; report hardware milestone slip.	Tapiwa
R-07	Scope exceeds four weeks	2 × 2 = 4	One water point; freeze mandatory scope; remove optional AI, SMS and solar work first.	Tapiwa / Fabilous
R-08	Errors in generated code or test evidence	2 × 3 = 6	Review generated code, run independent tests, label simulated evidence and implement physical-system software separately.	Fabilous / Claris / Lionel
R-09	Water damage or unsafe livestock trial	2 × 3 = 6	Protect connections, low-voltage supply and supervised small-container tests; defer live-animal deployment.	Tanatswa / Keith
R-10	User need or affordability is not confirmed	2 × 2 = 4	Review with an intended user; compare a mechanical float valve; record willingness to maintain and pay.	Tapiwa / Lionel

Stop wet testing immediately after an uncontrolled pump run, spill near electronics or failed stop function. Resume only after the defect is understood, repaired and the relevant safety test passes. Escalate material schedule or budget changes to the supervisor.

8 Throwaway prototype developed using vibe coding

8.1 Purpose and delivered artifact

TP 1.0 is an offline HTML simulation of the DDROID interface and control logic, packaged as DDROID_Throwaway_Prototype.html in DDROID_Prototype.zip. It runs in a browser and supports requirements evaluation before physical implementation.

The prototype starts stopped and demonstrates low/high refill behavior, supply shortage, sensor faults, high cutoff, manual fill, timeout, alarm acknowledgement, a demonstration role selector and event CSV export. All readings and events are synthetic. Percentage sliders generate scenarios; the physical specification uses discrete float states.

8.2 Vibe coding method and evidence

The prototype was developed through vibe coding: natural-language descriptions of user actions, control states, safety boundaries and offline behaviour guided AI-assisted code generation. The generated interface and isolated control logic were checked against the requirements. VIBE_CODING_RECORD.md documents the development brief and process; TEST_RESULTS.txt records 22 passing automated controller checks.

The completed checks cover simulated controller logic. Hardware response times, wired interlocks, authentication, persistent storage, interface usability and intended-user acceptance will be verified during physical implementation and the acceptance tests in section 9.

8.3 Review procedure

Review task	Question to resolve	Evidence to retain
Low trough and normal fill	Can the user see why the pump starts and stops?	Reviewer notes and labelled screenshot.
Low supply and sensor fault	Is the alarm and recovery sequence understood?	Observed action sequence and confusion points.
Manual fill and stop	Are controls easy to find and is restart deliberate?	Time to identify Stop and any erroneous action.
Records and link loss	Are simulated, stale and real measurements distinguishable?	Feedback linked to relevant requirement IDs.

8.4 Discard and transition decision

Following the Week 1 review, the team shall retain the resulting requirements, decisions, screenshots and development record, then archive the throwaway source as project evidence. The physical controller, local service and database shall be implemented separately against this SRS. The simulator’s role selector, in-memory log and artificial time steps are demonstration mechanisms.

Transition criteria are agreed screen labels, agreed normal and fault flows, assigned review actions, and supervisor review of the hardware scope.

9 Verification and requirements traceability

The following tests define acceptance of the physical system. Each test record shall include the actual result, evidence file, tester, date and any defect identifier. Acceptance requires every test to pass, verification of all associated requirements, and no unresolved failure affecting shutoff, access control or data integrity.

Test	Coverage	Acceptance evidence
AT-01	OBJ-02 FR-01 to 07 NFR-01 to 03	Run 20 refill cycles; verify switch truth table, debounce, hysteresis, response time and no overflow. Include a two-hour mixed run.
AT-02	OBJ-03 FR-04, 09, 11, 13 NFR-05 to 08	Test startup, power removal, wired stop and independent cutoff ten times, including halted application processing; reset remains disarmed.
AT-03	OBJ-03 FR-08 to 11 NFR-01	Inject low supply and inconsistent inputs; confirm pump stops, alarm latches, acknowledgement cannot clear it and reset is blocked until safe.
AT-04	OBJ-03 FR-07, 12	Block flow; verify maximum run. Verify manual fill stops at 15 seconds or earlier on FULL, fault or Stop.
AT-05	OBJ-03 FR-14, 15 NFR-12	Accept valid administrator setting only while stopped; reject invalid values and other roles; verify change log and restart persistence.
AT-06	OBJ-03 FR-16 to 18 NFR-04, 16	Validate herd count; write and filter records; export matching and empty CSVs; test deduplication, retention rotation and backup restore using seeded data.
AT-07	OBJ-03 FR-19, 20, 23 NFR-09 to 12	Test all roles with direct requests, disabled sessions, hashing, failed-login lockout, encryption, device identity and expired/repeated commands.
AT-08	OBJ-03 FR-21 to 23 NFR-02, 16	Disconnect Wi-Fi and gateway; verify local operation, stale flag, latest 100-event recovery, reported gaps and no delayed start replay.
AT-09	OBJ-01, 04 NFR-13 to 15	Five representative users complete stated tasks; check keyboard and phone layout; rebuild from recorded dependencies and run core tests.
AT-10	OBJ-05	Compare ten manual and ten automatic cycles under matched conditions; report time, overflow and interventions without assuming improvement.

The prototype’s 22 completed controller checks provide simulation evidence; physical-system acceptance is assessed separately through AT-01 to AT-10. Claris Chinanga maintains the acceptance record, supported by the technical leads responsible for test setup and defect resolution.

Appendix A Glossary

Term	Meaning
SRS	Software Requirements Specification: the agreed statement of required behavior and constraints.
Throwaway prototype	A temporary implementation used to learn and clarify requirements, then archived rather than adopted as the final system.
Vibe coding	Using natural-language instructions to assist code generation with AI, followed by review and testing.
Interlock	A condition or physical mechanism that inhibits a potentially unsafe operation.
Hysteresis	Separate low and high limits that prevent frequent switching around a single threshold.
Latched fault	A fault that remains active after its trigger clears until an explicit reset occurs.
IoT	Internet of Things; connected sensing and control. This baseline works on a local network without internet.
Telemetry	Status and measurement data sent by the control device.
Gateway	The local computer service that receives device messages and serves the dashboard and database.

Appendix B Analysis models

B1 Control state model

State	Pump	Transition rule
STOPPED	OFF	Valid arm enters IDLE or REFILLING; valid manual command enters MANUAL.
IDLE	OFF	Armed; LOW starts REFILLING. Stop disarms.
REFILLING	ON	FULL returns to IDLE; Stop returns to STOPPED; interlock enters FAULT.
MANUAL	ON	Full, 15-second expiry or Stop returns to STOPPED; interlock enters FAULT.
FAULT	OFF	Correct cause and reset to STOPPED; a separate arm is needed to restart.

Safety precedence applies in every state. Physical cutoff can remove pump drive regardless of the software state. Loss of dashboard communication does not change local control state. The analysis model is implemented in simplified form in TP 1.0.

Appendix C References

The following sources support the project context, requirements structure and technical design.

[R1] Software Requirements Specification template

Coursework template, supplied as srs_template-ieee (1).doc.

[R2] Group Project coursework brief

Project instructions and assessment requirements for the Software Requirements Specification, group documentation and throwaway prototype developed using vibe coding.

[R3] Food and Agriculture Organization of the United Nations

Weir dams bring hope to rural communities struggling with water scarcity, 10 January 2025. Accessed 12 October 2026.
https://www.fao.org/africa/news-stories/news-detail/weir-dams-bring-hope-to-rural-communities-struggling-with-water-scarcity/

[R4] Espressif Systems

ESP32 Series Datasheet. Accessed 12 October 2026.
https://documentation.espressif.com/esp32_datasheet_en.html

[R5] OWASP Foundation

Password Storage Cheat Sheet. Accessed 12 October 2026.
https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html